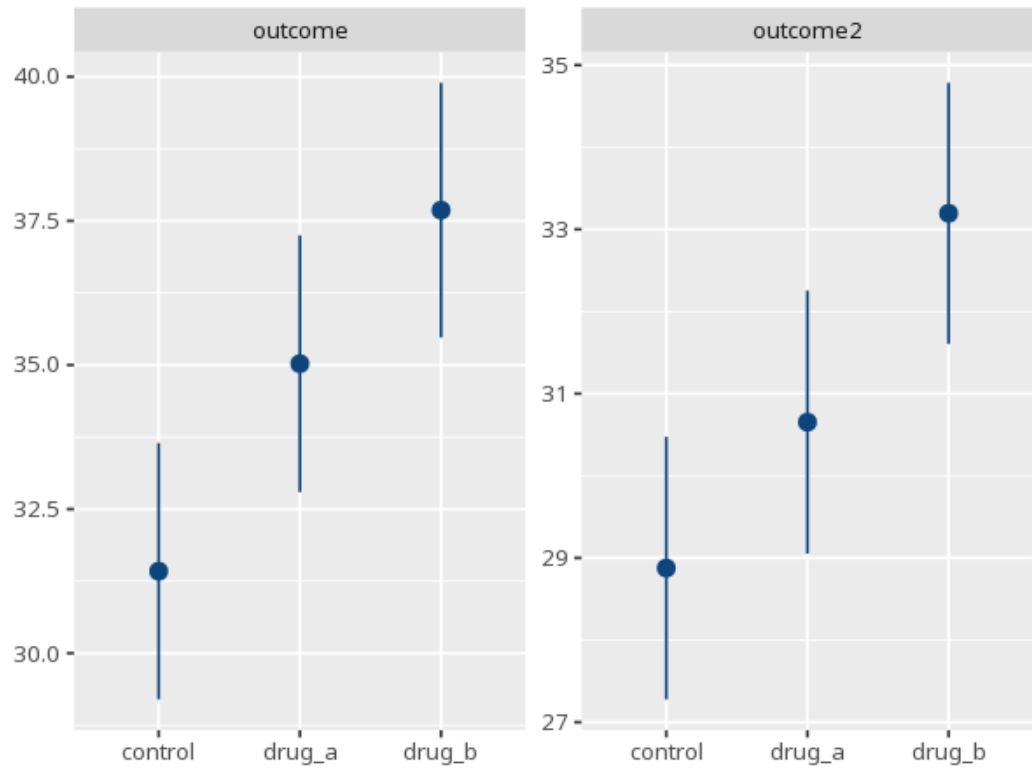


1 MANCOVA

Effect	Pillai / Wilks	approx F	df1	df2	p	partial η^2 [95% CI]
baseline	0.55	33.87	2	55	<.001	0.55 [0.40, 1.00]
group	0.37	6.30	4	112	<.001	0.18 [0.07, 1.00]

DV	F	df1	df2	p	partial η^2 [95% CI]
outcome	8.07	2	56	<.001	0.22 [0.07, 1.00]
outcome2	7.47	2	56	.001	0.21 [0.06, 1.00]



Like MANOVA, but group differences on the set of outcomes are assessed after controlling for one or more covariates. Interpret the univariate follow-ups only if the multivariate p is significant, and adjust them for the number of DVs (e.g. Bonferroni) to control familywise error. Your significance threshold (α) is 0.05.

A MANCOVA gave a covariate-adjusted group effect, Pillai's $V=.37$, $F(4,112)=6.30$, $p < .001$, partial $\eta^2=.18$ [.07, 1.00].

Statistical basis: Pillai's trace multivariate test (headline statistic, covariate-adjusted) (Pillai, K. C. S. (1955). "Some new test criteria in multivariate analysis." *Annals of Mathematical Statistics*, 26(1), 117-121.); Wilks' Lambda multivariate test (covariate-adjusted) (Wilks, S. S. (1932). "Certain generalized

distributions in multivariate analysis." *Biometrika*, 24(3/4), 471-494.); Box's M test of homogeneity of covariance matrices (Box, G. E. P. (1949). "A general distribution theory for a class of likelihood criteria." *Biometrika*, 36(3-4), 317-346.); Multivariate effect size (partial η^2 from the multivariate approx. F) (Ben-Shachar MS, Lüdtke D, Makowski D (2020). "effectsize: Estimation of Effect Size Indices and Standardized Parameters." *Journal of Open Source Software*, 5(56), 2815.); Estimated marginal (covariate-adjusted) means (Lenth RV (2024). *emmeans: Estimated Marginal Means, aka Least-Squares Means*. R package.).